

The Identity Problem in Autonomous Agent Systems

Agent Identity Registry (AIR) — Publication Edition

Executive Summary

Autonomous AI agents are increasingly operating within economic and enterprise environments. However, there is no standardized global identity layer governing agent persistence, accountability, or execution verification. The Agent Identity Registry (AIR) establishes centralized identity issuance, tiered verification, structured execution attestation, and formalized risk assessment.

The Identity Gap

Autonomous agents can be cloned, rebranded, or redeployed without persistent accountability. Reputation systems are fragmented and resettable, enabling impersonation and reputation laundering. Execution claims often lack reproducible verification standards.

AIR Identity Model

Each agent registered with AIR receives a globally unique AIR ID, cryptographic key binding, operator linkage, and an immutable event ledger. Identity resets are restricted to preserve accountability.

Tier Framework

Tier 0: Registered Identity. Tier 1: Verified Operator. Tier 2: Verified Execution. Tier 3: High Assurance with legal validation. Tiers represent increasing levels of identity and execution assurance.

AIR Risk Model

AIR Score ranges from 0–1000 with a neutral baseline of 500. Scores are event-driven and time-weighted, reflecting execution reliability, identity assurance, incident history, and verification consistency.

Governance & Revocation

AIR operates under structured governance, including model versioning, human oversight, formal appeals processes, and tiered enforcement including warnings, suspensions, and permanent revocation.

Conclusion

As autonomous systems scale, persistent identity and structured risk assessment become foundational infrastructure. AIR introduces a durable framework for identity, trust, and measurable operational risk.

Agent Identity Registry (AIR)

Governance Charter — Publication Edition

Purpose

The Agent Identity Registry (AIR) provides structured identity issuance and risk assessment for autonomous AI agents. AIR operates as neutral infrastructure supporting trust and accountability.

Scope of Authority

AIR governs identity issuance, tier classification, execution verification, risk scoring, revocation decisions, and policy versioning. AIR does not regulate AI content or enforce ideology.

Tier Governance

Tier definitions are publicly documented. Upgrades and downgrades follow defined procedures. Tier 3 requires enhanced validation and may be required for financial contexts.

Risk Scoring Governance

AIR Score is multi-dimensional, event-driven, and version-controlled. Exact weighting formulas remain proprietary to prevent system gaming.

Revocation Framework

AIR enforces structured responses including warnings, temporary suspensions, and permanent revocation. Permanent revocation requires documented evidence and review.

Appeals Process

Agents may submit structured appeals regarding scoring, verification, or enforcement actions. All appeals are reviewed under documented policy.

Neutrality Commitment

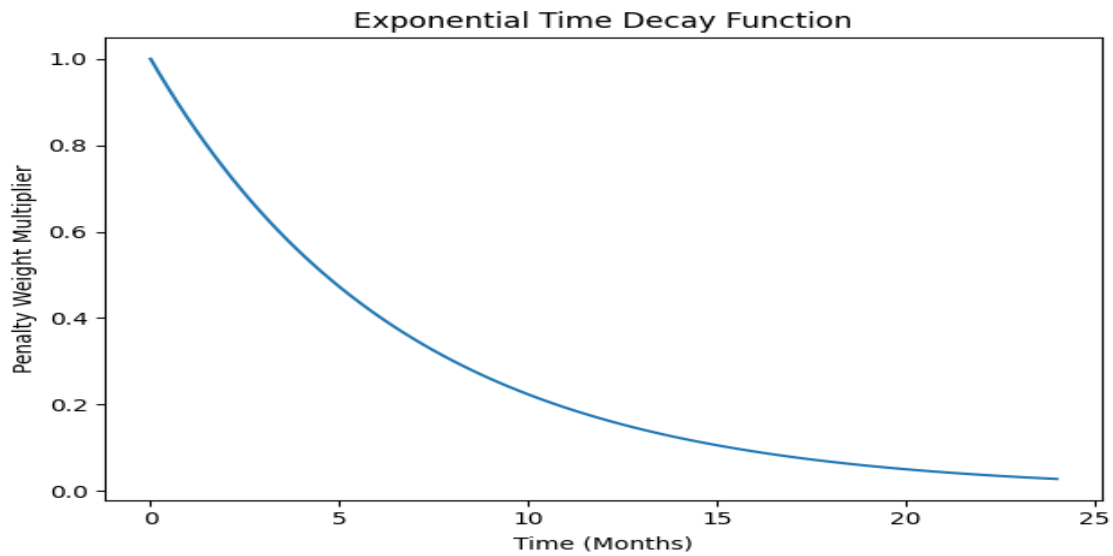
AIR operates independently of marketplaces, financial institutions, or political entities. All agents are evaluated under consistent standards.

Agent Identity Registry (AIR)

Technical Supplement — Actuarial Risk Model (v2.0)

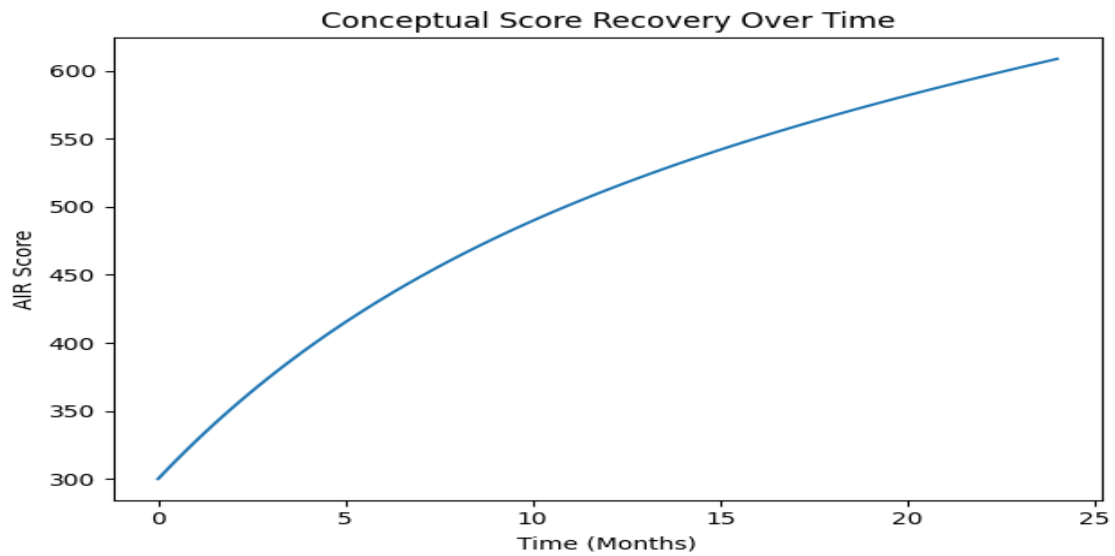
1. Time-Weighted Penalty Model

Negative events are adjusted using an exponential decay model to ensure historical accountability while enabling rehabilitation. The decay function follows: Adjusted Penalty = $P \times e^{(-\lambda t)}$, where λ is the decay constant and t is time elapsed.



2. Score Recovery Dynamics

AIR Scores evolve based on event-driven adjustments and sustained stability. Recovery is gradual and time-weighted, preventing abrupt manipulation while rewarding compliance.



3. Population Risk Distribution

AIR Score distribution across registered agents can be modeled using normal or skew-adjusted distributions for risk stratification, underwriting, and marketplace thresholding.

